

FinSight: An Agentic AI Based Financial Research & Analysis System

AD23731 Foundations of Agentic AI - Mini-Project

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in partial fulfilment of the requirements for the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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RAJALAKSHMI ENGINEERING COLLEGE

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OCTOBER 2026



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BONAFIDE CERTIFICATE

Certified that this mini-project report “**FinSight — An Agentic AI Based Financial Research & Analysis System**” is the bonafide work of Kishanthraja M (2116231501077), Mervin Anto Santhosh A (2116231501099), Raguchanthar S (2116231501126) VII Semester, B.Tech. Artificial Intelligence and Machine Learning Department in partial fulfilment of the requirements of the course **AD23731 – Foundations of Agentic AI** during the Academic Year 2026 – 2027.

Faculty in-charge

Submitted to mini-project viva voce examination held on:

INTERNAL EXAMINER

EXTERNAL EXAMINER

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ABSTRACT

Financial research requires the collection and analysis of information from multiple sources, including financial reports, company filings, market data, regulatory disclosures, and financial news. Conventional manual research is time-consuming and makes it difficult to efficiently combine information from heterogeneous sources. Although Large Language Models (LLMs) and basic Retrieval-Augmented Generation (RAG) systems can assist in financial analysis, they may produce incomplete or unsupported insights when retrieval is fixed and does not adapt to the research requirement. To address these limitations, this project proposes FinSight, an Agentic AI Based Financial Research and Analysis System that automates the financial research process through a coordinated multi-agent architecture.

FinSight employs specialised agents for different stages of financial research. A Planner Agent decomposes the user's financial query into research tasks, while the Financial Research Agent, Market Data Agent, and News and Sentiment Agent retrieve and analyse relevant information from financial documents, market-data APIs, and news sources. A Risk Analysis Agent identifies financial, market, competitive, and regulatory risks. The findings are then consolidated by a Synthesis Agent into a structured financial research report. A Critic Agent validates the retrieved evidence, checks for contradictions, evaluates coverage, and identifies unsupported claims. The system incorporates Agentic RAG, where retrieval queries are dynamically generated and refined when the available evidence is insufficient, enabling more adaptive and evidence-driven research.

The proposed system uses financial reports and company filings from sources such as SEC EDGAR and company investor-relations portals, market information from Yahoo Finance and Alpha Vantage, and relevant financial news and industry sources. LangGraph is used for multi-agent workflow orchestration and state management, while LangChain supports retrieval and tool integration. FAISS/ChromaDB and Sentence Transformers support document retrieval and embeddings.

FinSight further incorporates a Human-in-the-Loop (HITL) mechanism, allowing users to review, approve, reject, or request re-analysis of AI-generated findings. Rejected findings trigger targeted re-planning and retrieval rather than restarting the complete workflow. The expected outcome is an evidence-backed financial research system capable of producing structured insights, market analysis, risk assessment, supporting evidence, and traceable research reports while improving reliability, adaptability, and transparency in AI-assisted financial analysis

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I. INTRODUCTION

1.1 Background and motivation

Financial research involves collecting and analysing information from multiple sources, including financial reports, company filings, market data, regulatory disclosures, and financial news. Analysts and investors often need to combine information from these heterogeneous sources to understand a company's financial performance, market position, opportunities, and potential risks. However, conventional financial research is largely manual and time-consuming, particularly when information must be collected, compared, and validated across different sources.

The emergence of Large Language Models (LLMs) has introduced new possibilities for automating financial research. LLMs can understand natural-language questions, summarize documents, and generate analytical responses. However, a conventional LLM may generate information that is incomplete, outdated, or unsupported by reliable evidence. Basic Retrieval-Augmented Generation (RAG) systems improve grounding by retrieving relevant documents before generating an answer, but their retrieval process is generally fixed and may not adapt to the specific requirements of a complex financial research task.

FinSight addresses this limitation through an Agentic AI-based approach. Instead of relying on a single AI model, the system uses multiple specialised agents that collaborate through an orchestrated workflow. A Planner Agent first understands the user's research question and decomposes it into smaller tasks. Dedicated agents then retrieve and analyse financial information, market data, news, and risk factors. Their findings are subsequently combined and validated before producing the final research report.

1.2 Need for Agentic AI

The complexity of financial research makes it suitable for an agent-based approach because different research activities require different sources, tools, and reasoning processes. FinSight therefore separates financial research into specialised tasks rather than asking a single LLM to perform the entire analysis. The system dynamically determines what evidence is required and can perform iterative retrieval when the initial evidence is insufficient. This enables the retrieval process to adapt to the research requirement rather than following a fixed retrieval strategy.

The system also incorporates Human-in-the-Loop (HITL) validation to improve reliability. AI-generated findings are presented with confidence levels and their

supporting evidence. A human user can approve or reject a finding and provide feedback. When a finding is rejected, system performs targeted re-planning and re-analysis of the affected topic instead of repeating the entire workflow.

1.3 Project Objectives

The primary objective of FinSight is to develop an intelligent financial research and analysis system capable of producing evidence-backed financial insights. The specific objectives are:

1. To develop a multi-agent AI architecture for automated financial research.
2. To decompose complex financial queries into smaller research tasks.
3. To retrieve relevant information from financial reports, market data, and news sources.
4. To implement Agentic RAG with adaptive and iterative retrieval.
5. To perform financial, market, sentiment, and risk analysis using specialised agents.
6. To validate generated findings using a dedicated Critic Agent.
7. To incorporate Human-in-the-Loop validation and targeted re-analysis.
8. To generate structured financial research reports with supporting evidence and sources.

The overall objective is to provide financial researchers, investors, decision makers, and students with a system that can efficiently transform a financial research question into a structured, reliable, and evidence-backed analysis

II. PROBLEM STATEMENT AND BUSINESS CASE

2.1 Problem Definition

Financial research requires information to be collected from multiple heterogeneous sources such as financial reports, market data, company filings, regulatory disclosures, and financial news. Analysing these sources manually is time-consuming and makes it difficult to efficiently combine, compare, and interpret information from different sources.

Traditional Large Language Model (LLM) applications can generate financial explanations and summaries, but they may produce incomplete or unsupported insights. Similarly, basic Retrieval-Augmented Generation (RAG) systems generally depend on a fixed retrieval process and may not dynamically adapt to the requirements of a complex research question. This can result in insufficient evidence and reduced reliability of the generated analysis.

Therefore, there is a need for an intelligent financial research system that can understand a user's query, plan the required research tasks, retrieve relevant evidence, analyse financial information, validate the findings, and generate a reliable research report with human oversight.

2.2 Business and Real-World Context

Financial researchers and analysts need to quickly gather information about companies and markets, while investors and decision makers require an understanding of financial performance, opportunities, and potential risks. Students and researchers also require structured methods for conducting financial research using multiple reliable sources.

In a real-world financial research workflow, information may be distributed across company filings, financial statements, market-data platforms, news reports, and regulatory sources. Manually collecting and comparing this information increases research effort and can make it difficult to maintain a consistent and evidence-based analysis.

FinSight addresses this challenge by providing a **multi-agent financial research workflow**. The Planner Agent decomposes the research question, while specialised agents independently gather financial, market, news, and risk-related evidence. These results are combined by the Synthesis Agent and subsequently checked by the Critic Agent.

The proposed system is intended for **financial researchers/analysts, investors/decision makers, and students/researchers**. Its expected outcome is an evidence-backed financial research report containing financial insights, market information, risk analysis, and supporting sources.

2.3 Business Need

The major business needs addressed by FinSight are:

- **Reduce research time** by automating information gathering and analysis.

- **Integrate multiple information sources** into a single research workflow.
- **Improve evidence grounding** through Agentic RAG and source-based retrieval.
- **Identify financial and market risks** systematically.
- **Reduce unsupported AI-generated claims** through Critic Agent validation.
- **Enable human oversight** for important or uncertain findings.
- **Generate structured research reports** instead of isolated AI responses.

Thus, FinSight aims to transform financial research from a largely manual and fragmented process into an **adaptive, multi-agent, evidence-driven workflow**.

III. USER AND STAKEHOLDER ANALYSIS

3.1 Target Users

FinSight is designed for users who require efficient and structured financial research from multiple reliable sources. The primary target users are financial researchers and analysts, investors and decision makers, and students and researchers.

Financial Researchers / Analysts

Financial researchers and analysts can use FinSight to quickly gather and analyse information about companies and financial markets. The system automates the collection of financial reports, market information, company data, and relevant news, reducing the effort required for manual research.

Investors / Decision Makers

Investors and decision makers require information about financial performance, opportunities, and potential risks before making informed decisions. FinSight provides structured analysis by combining financial data, market information, news sentiment, and risk factors.

Students / Researchers

Students and academic researchers can use FinSight to perform structured financial research using information collected from multiple reliable sources. The

system can help them organise evidence and generate a comprehensive research output rather than manually analysing each source independently.

3.2 Stakeholder Identification

TABLE I
STAKEHOLDERS AND THEIR INTERESTS

The major stakeholders of the proposed system are:

Stakeholder	Requirement / Interest	FinSight's Response
Financial Researchers / Analysts	Quickly gather and analyse company and market information	Automated multi-agent research and analysis
Investors / Decision Makers	Understand performance, opportunities and risks	Financial, market and risk analysis
Students / Researchers	Conduct structured research using reliable sources	Evidence-backed research reports
System Users	Reliable and understandable findings	Source-supported insights and structured reports
Project Evaluators	Demonstration of an intelligent and reliable AI system	Agentic workflow, Agentic RAG and validation mechanisms

The project specifically identifies financial researchers/analysts, investors/decision makers, and students/researchers as the key user groups.

3.3 User Requirements

Based on the proposed FinSight workflow, the system should satisfy the following requirements:

Functional Requirements

1. **Query Input** – The system should accept a user's financial research question.
2. **Task Decomposition** – The Planner Agent should break the query into appropriate research tasks.
3. **Financial Research** – The system should retrieve and analyse company reports, filings, and financial information.
4. **Market Analysis** – The Market Data Agent should retrieve market data and perform relevant calculations.
5. **News Analysis** – The News and Sentiment Agent should retrieve recent news and analyse market sentiment.

6. **Risk Analysis** – The Risk Analysis Agent should identify financial, market, competitive, and regulatory risks.
7. **Evidence Retrieval** – The system should retrieve relevant evidence from its knowledge sources.
8. **Iterative Retrieval** – The system should refine the retrieval query when the available evidence is insufficient. Finsight Final
9. **Synthesis** – The Synthesis Agent should combine the outputs of the different agents into a coherent report.
10. **Validation** – The Critic Agent should identify unsupported claims, contradictions, and incomplete findings.
11. **Human Review** – Users should be able to approve or reject AI-generated findings.
12. **Re-analysis** – Rejected findings should trigger targeted retrieval and analysis of the affected topic. Finsight Final
13. **Report Generation** – The system should produce an evidence-backed financial research report.

Non-Functional Requirements

- **Reliability:** Findings should be supported by relevant evidence.
- **Adaptability:** Retrieval should dynamically respond to the research requirement.
- **Traceability:** Findings should be associated with their supporting sources.
- **Usability:** The system should provide a clear interface for submitting queries and reviewing results.
- **Efficiency:** Only relevant agents should be re-executed when corrections are required.
- **Human Oversight:** Users should retain control over important findings before final report generation.

IV. PROPOSED AGENTIC AI SOLUTION

A. Overall Solution

FinSight is proposed as a **multi-agent AI system for automated financial research and analysis**. The system accepts a user's financial research query and

uses a coordinated set of specialised agents to plan the research, retrieve relevant evidence, analyse financial information, identify risks, validate findings, and generate a structured financial research report.

Instead of depending on a single LLM to perform the complete task, FinSight divides the research process into specialised activities. A **Planner Agent** first understands the user's query and decomposes it into research tasks. The corresponding agents then perform financial research, market-data analysis, news and sentiment analysis, and risk analysis. Their outputs are passed to the Synthesis Agent, which produces a coherent analysis. Finally, the Critic Agent verifies the evidence and identifies unsupported or incomplete findings.

B. Key Features

1. Automated Research Planning

The **Planner Agent** understands the user's financial question and breaks it into smaller research tasks. This allows different aspects of a complex query to be handled by specialised agents.

2. Multi-Agent Financial Research

FinSight contains specialised agents for:

- Financial reports and company information
- Market and stock data
- News and sentiment
- Financial and regulatory risk
- Report synthesis
- Evidence validation

3. Agentic RAG

The system does not rely on a single fixed retrieval query. The Financial Research Agent determines what evidence is required, generates targeted retrieval queries, evaluates the retrieved information, and performs **iterative retrieval** when the evidence is insufficient.

4. Evidence-Based Analysis

Financial reports, company filings, market data, news, and other knowledge sources are used to ground the generated findings. This helps reduce unsupported financial claims.

5. Risk Analysis

The **Risk Analysis Agent** identifies different categories of risks, including:

- Financial risks
- Market risks
- Competitive risks
- Regulatory risks

6. Critic-Based Validation

The **Critic Agent** checks whether the generated findings have adequate evidence and identifies contradictions, incomplete information, and unsupported claims before the final report is produced.

7. Human-in-the-Loop

Users can review AI-generated findings and either **approve or reject** them. If a finding is rejected, the system performs targeted re-planning and re-analysis instead of restarting the complete workflow.

C. Agent-Based Approach

The proposed solution assigns a specific responsibility to each agent:

Agent	Responsibility
Planner Agent	Understands the query and creates research tasks
Financial Research Agent	Retrieves and analyses financial reports, filings and company information
Market Data Agent	Retrieves market data and performs calculations
News & Sentiment Agent	Retrieves news and analyses market sentiment
Risk Analysis Agent	Identifies financial, market, competitive and regulatory risks
Synthesis Agent	Combines findings into a structured report
Critic Agent	Validates evidence and identifies unsupported findings

The agents are coordinated rather than operating independently. Their outputs are passed through the workflow so that the findings from financial research, market data, news, and risk analysis can be combined into a single evidence-backed financial analysis.

V. DATASET AND KNOWLEDGE SOURCES

A. Dataset or Knowledge Source Description

FinSight is designed as a **knowledge-driven financial research system** that retrieves information from multiple financial and market sources rather than depending on a single dataset. The knowledge sources provide the evidence required by the different agents during the financial research process.

The major knowledge sources identified for the system are:

1. Financial Reports and Company Filings

Financial reports and company filings provide the primary information required for analysing a company's financial condition and performance.

Sources:

- **SEC EDGAR**
- **Company Investor Relations portals**

Information includes:

- Annual reports
- Quarterly reports
- Financial statements
- Company disclosures

These sources are primarily used by the **Financial Research Agent** to retrieve and analyse company-specific information.

2. Market and Financial Data

Market-data sources provide quantitative information required for analysing market performance.

Sources:

- **Yahoo Finance**
- **Alpha Vantage APIs**

The data may include:

- Historical stock prices
- Trading volume
- Financial indicators
- Other relevant market information

The **Market Data Agent** uses these sources to retrieve information and perform relevant calculations.

3. Financial News and Industry Reports

Recent financial developments cannot always be obtained from historical financial statements. Therefore, FinSight also considers:

- Financial news
- Industry reports

4. Regulatory Disclosures

Regulatory disclosures provide additional information that may be relevant for identifying **regulatory and financial risks**. These sources are considered by the research and risk-analysis components of the system.

VI. TOOLS AND TECHNOLOGIES USED

TABLE III
TECHNOLOGY STACK

Category	Technology / Tool	Purpose in FinSight
Programming Language	Python	Core implementation of agents, data processing and backend logic
AI Model	GPT-4 / Gemini / Claude	Reasoning, analysis, generation and report synthesis
Agent Framework	LangGraph	Multi-agent orchestration, workflow control, state management and HITL checkpoints
AI Framework	LangChain	RAG, retrievers and tool integration
Embeddings	Sentence Transformers	Converts financial document chunks into vector embeddings
Vector Database	FAISS / ChromaDB	Stores and retrieves financial document embeddings
Database	SQLite / PostgreSQL	Stores user queries, workflow state and report history
Financial Data	Yahoo Finance / Alpha Vantage	Historical prices, volume and financial indicators
Financial Documents	SEC EDGAR / Company Filings	Annual reports, quarterly reports and financial statements
News Sources	News / Search APIs	Recent financial news and industry developments
Backend	FastAPI	Backend API and agent orchestration services
Frontend	Streamlit	Interactive interface for financial research queries and reports
Visualization	Plotly	Interactive market charts and financial trends
Evaluation Visualization	Matplotlib	Evaluation graphs such as accuracy, hallucination rate and retrieval-cost comparisons

VII. AGENT ARCHITECTURE

The system is organised in layers, shown in Fig. 1: a client layer (dashboards, Swagger UI, and the command-line interface), a FastAPI service layer, the LangGraph orchestration layer that hosts the seven agents, supporting knowledge, execution, and research services, and a persistence layer.

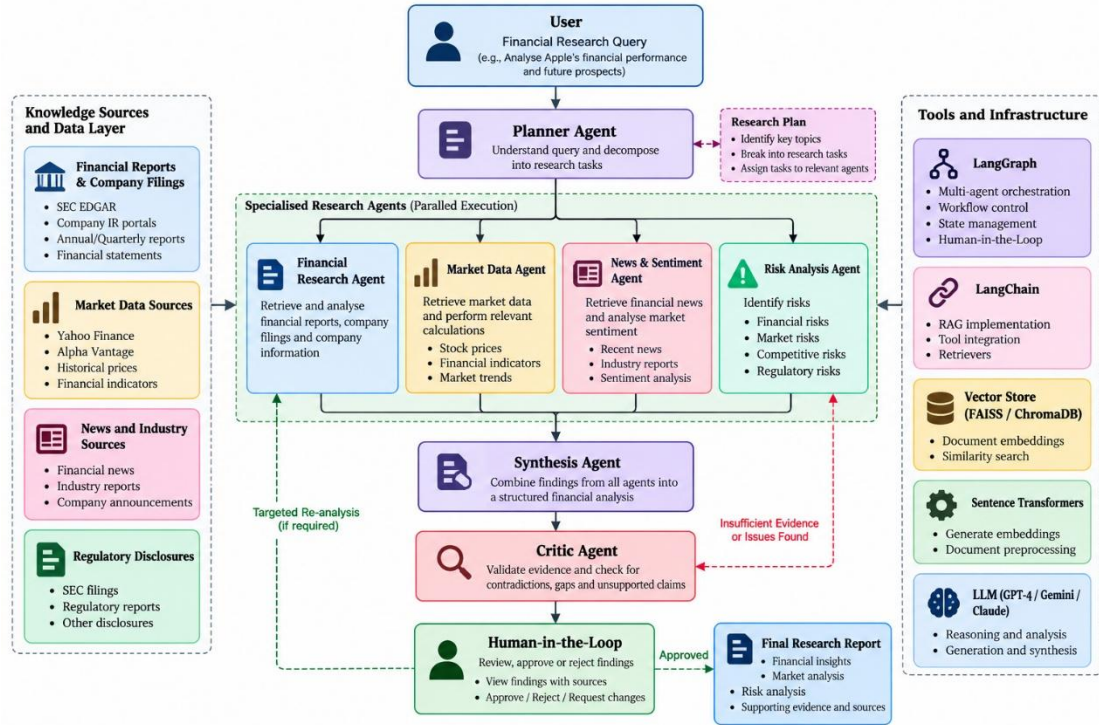


Fig. 1. Layered architecture

A. Agent Tools

TABLE V
AGENT ROLES AND DELIVERABLES*

Agent	Tools / Data Used
Planner Agent	LLM + workflow state
Financial Research Agent	Financial reports, company filings, vector database
Market Data Agent	Yahoo Finance / Alpha Vantage APIs
News & Sentiment Agent	News and search APIs
Risk Analysis Agent	Retrieved financial evidence + research findings
Synthesis Agent	Outputs from all research agents
Critic Agent	Retrieved evidence + generated findings

The proposed framework uses **LangChain** for RAG and tool integration and **LangGraph** for multi-agent orchestration and state management.

B. Agent components

Agent	Role	Primary Responsibility
Planner Agent	Research Planner	Understands the user's query and decomposes it into research tasks
Financial Research Agent	Financial Analyst	Retrieves and analyses financial reports, company filings and company information
Market Data Agent	Market Analyst	Retrieves market data and performs relevant calculations
News & Sentiment Agent	News Analyst	Retrieves relevant financial news and analyses market sentiment
Risk Analysis Agent	Risk Analyst	Identifies financial, market, competitive and regulatory risks
Synthesis Agent	Report Generator	Combines findings from different agents into a structured financial research report
Critic Agent	Validator	Checks evidence, contradictions, coverage and unsupported findings

C. Memory

The system maintains information about the ongoing research workflow so that outputs from one stage can be passed to subsequent stages. The workflow state contains information such as:

- User query
- Research tasks
- Retrieved evidence
- Agent findings
- Risk analysis
- Synthesis results
- Critic feedback
- Human validation decisions

This shared state allows the agents to work as a coordinated

D. Reasoning Flow

Within the workflow each agent follows a plan, act, observe, and record cycle. The agent receives the relevant slice of global state, reasons about the task, invokes its tools, validates its output against its own criteria (for example, unit tests pass or a citation exists for each standards claim), and records input, output, citations, latency, and issues in `agent_runs`. Where validation fails, the agent reports issues to the orchestrator rather than silently continuing, which enables reflection.

E. Agent Interactions

The Planner determines what needs to be researched, the specialised agents determine what the evidence says, the Risk Agent determines what could go wrong, the Synthesis Agent determines how the findings should be combined, and the Critic Agent determines whether the resulting analysis is adequately supported.

This architecture enables FinSight to transform a single financial question into a structured, evidence-backed and validated financial research report.

VIII. AGENT WORKFLOW AND ORCHESTRATION

A. Workflow Design

FinSight follows a **LangGraph-based multi-agent workflow** in which the user's financial research query moves through a sequence of specialised agents. The workflow begins with planning, continues through parallel evidence collection and analysis, and ends with synthesis, validation, human review, and final report generation.

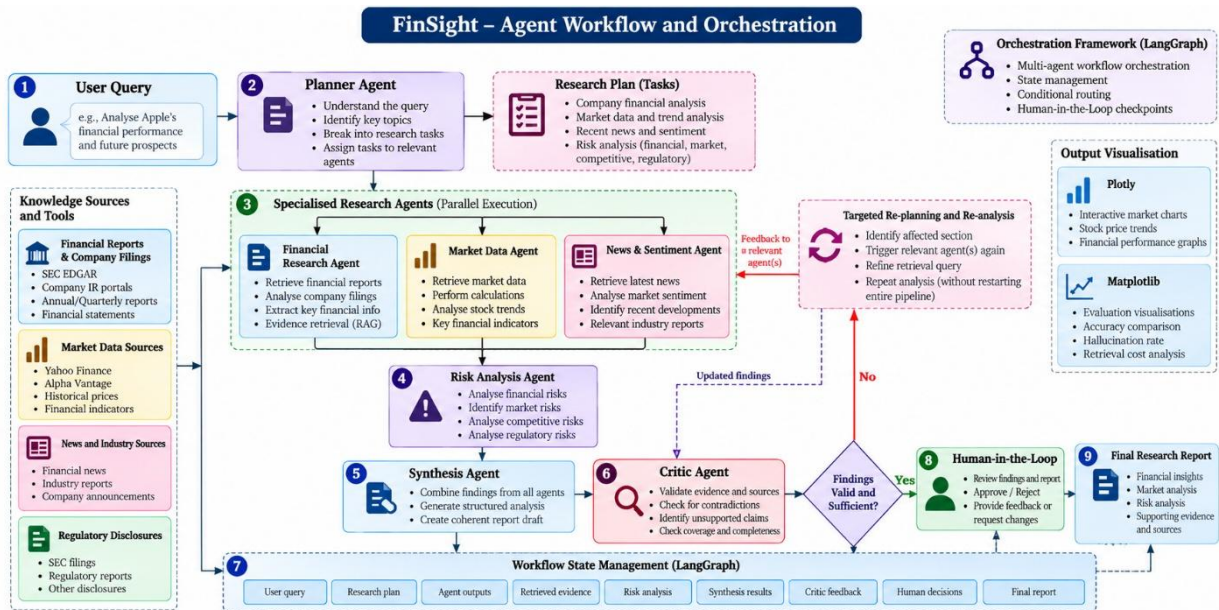


Fig. 2. Agent workflow

B. Task Decomposition

The **Planner Agent** receives the user's financial query and decomposes it into smaller research tasks. These tasks are then assigned to the appropriate specialised agents.

Stage	Agent / Process	Output
1	User Query	Financial research question
2	Planner Agent	Research plan and task decomposition
3	Financial Research Agent	Financial reports, filings and company evidence
4	Market Data Agent	Market data and financial calculations
5	News & Sentiment Agent	Recent news and sentiment insights
6	Risk Analysis Agent	Financial, market, competitive and regulatory risks
7	Synthesis Agent	Combined financial analysis
8	Critic Agent	Evidence validation and issue identification
9	Human-in-the-Loop	Approval, rejection or feedback
10	Final Output	Evidence-backed financial research report

C. Agent Coordination

The research agents operate in a coordinated manner. The Financial Research Agent, Market Data Agent, and News & Sentiment Agent can gather evidence in parallel, which reduces unnecessary waiting between independent research tasks. Their findings are then passed to the Risk Analysis Agent and Synthesis Agent.

The Synthesis Agent combines the outputs into a coherent analysis, while the Critic Agent checks the completeness and reliability of the generated findings.

If the Critic Agent identifies insufficient evidence, contradictions, or unsupported claims, the workflow can return to the relevant retrieval or analysis stage instead of immediately producing the final report.

D. Orchestration Implementation

LangGraph is proposed as the orchestration framework for controlling the FinSight workflow. It manages:

- Agent execution order
- Shared workflow state

- Conditional routing
- Feedback loops
- Human-in-the-Loop checkpoints

The shared workflow state allows outputs generated by one agent to be used by subsequent agents. This makes it possible to preserve the research plan, retrieved evidence, analysis results, validation feedback, and human decisions throughout the execution process.

IX. AGENTIC RAG IMPLEMENTATION

A. Retrieval Architecture

FinSight implements **Agentic Retrieval-Augmented Generation (Agentic RAG)** to provide reliable and evidence-backed financial analysis. Instead of allowing the LLM to generate answers only from its internal knowledge, the system retrieves relevant information from financial documents, company filings, market sources, and news before generating the final analysis.

B. Knowledge Retrieval

FinSight uses multiple knowledge sources to retrieve relevant financial information, including:

- **SEC EDGAR**
- **Company Investor Relations portals**
- **Annual and quarterly reports**
- **Financial statements**
- **Yahoo Finance**
- **Alpha Vantage**
- **Financial news**
- **Industry reports**
- **Regulatory disclosures**

Financial documents are processed into smaller text chunks and converted into vector embeddings using **Sentence Transformers**. These embeddings are stored in a vector database such as **FAISS or ChromaDB**, allowing semantically relevant information to be retrieved for a given research task.

C. Generation Process

Once sufficient evidence has been retrieved, the relevant agents use that evidence to perform their respective tasks.

For example:

- **Financial Research Agent** → analyses financial reports and company information.
- **Market Data Agent** → analyses prices, volume and financial indicators.
- **News & Sentiment Agent** → analyses recent news and sentiment.
- **Risk Analysis Agent** → identifies financial, market, competitive and regulatory risks.

The **Synthesis Agent** then combines these findings into a structured financial analysis. The generated findings are subsequently passed to the **Critic Agent**, which checks the evidence and identifies contradictions, unsupported claims, or missing information.

D. Retrieval-Quality Evaluation

FinSight evaluates the effectiveness of Agentic RAG by comparing three configurations:

TABLE VII

Configuration	Retrieval Approach	Retrieval Evaluation
LLM-only	No retrieval	Baseline
Basic RAG	Fixed/single-stage retrieval	Compared against proposed system
Agentic RAG	Adaptive retrieval with iterative refinement	Proposed approach

The project evaluates the RAG approach using metrics such as:

- **Accuracy**
- **Hallucination rate**
- **Retrieval cost**
- **Evidence sufficiency**

The objective is to demonstrate that adaptive Agentic RAG can provide more reliable evidence for financial analysis than a conventional LLM-only or basic RAG approach.

X. SYSTEM IMPLEMENTATION

A. Functional Prototype Screenshots

The prototype is a backend service with an interactive API console; its principal interfaces are shown in the following figures.

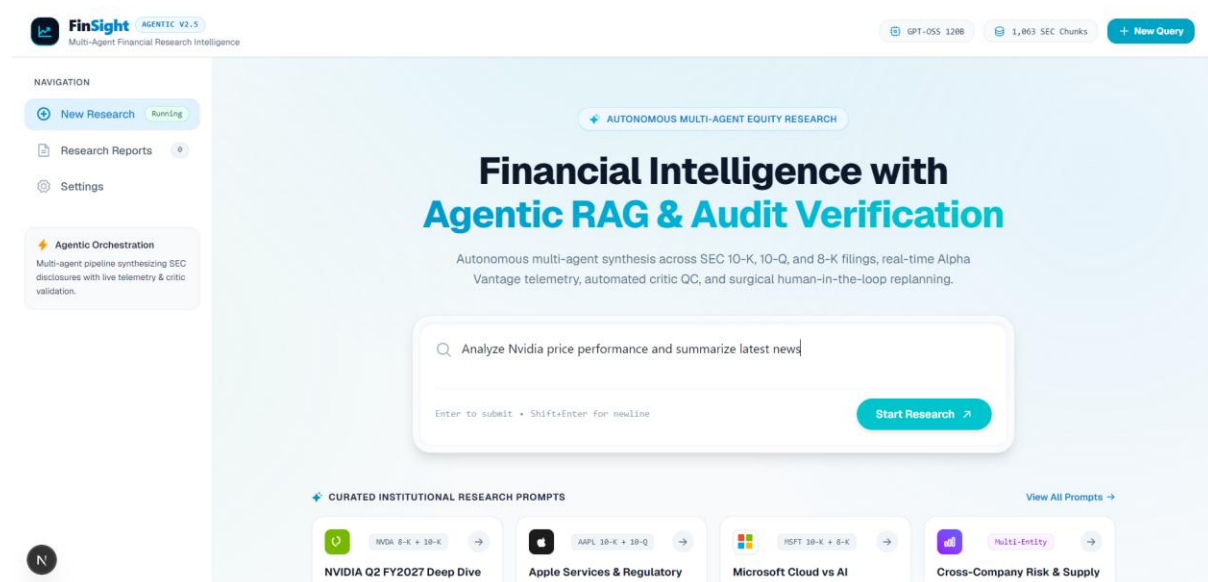


Figure 10.1. Research Query Dashboard

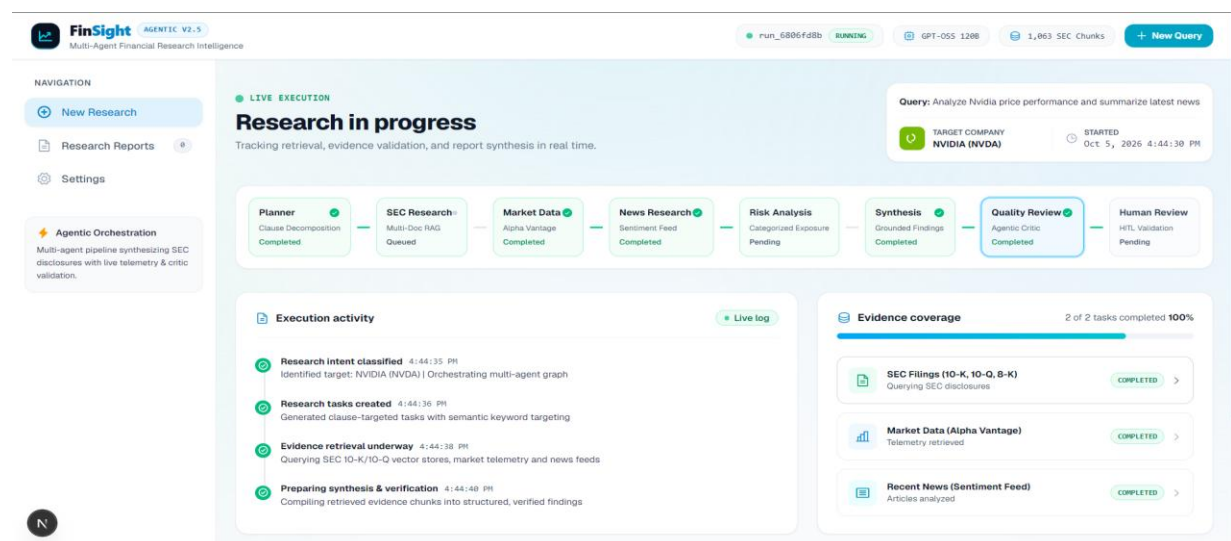


Figure 10.2. Multi-Agent Research Execution Dashboard

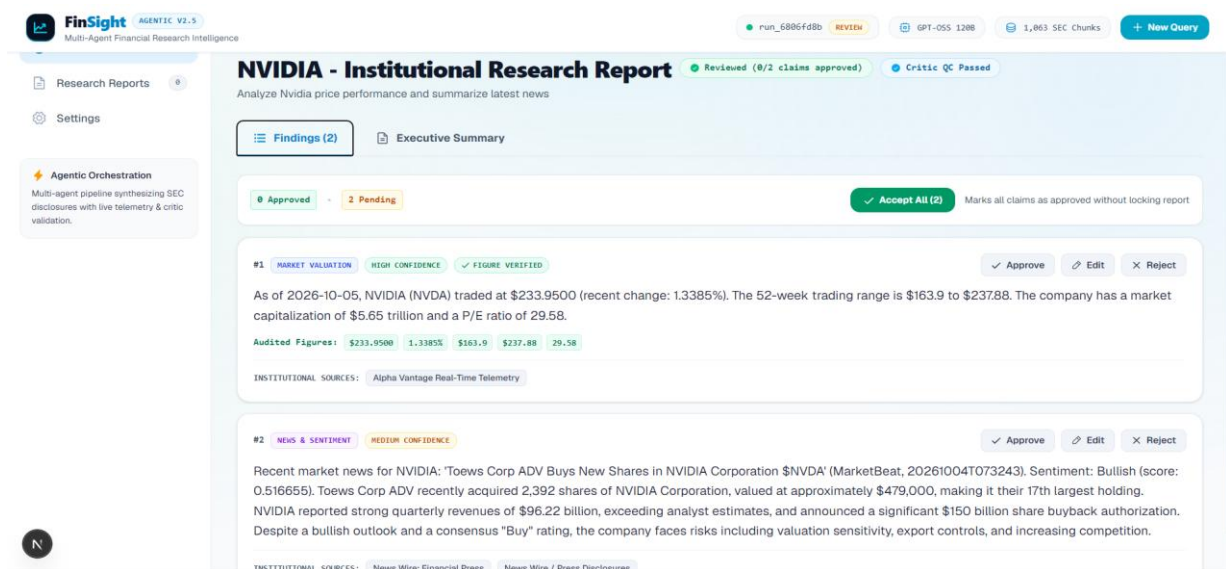


Figure 10.3. FinSight Human-in-the-Loop Research Audit Console

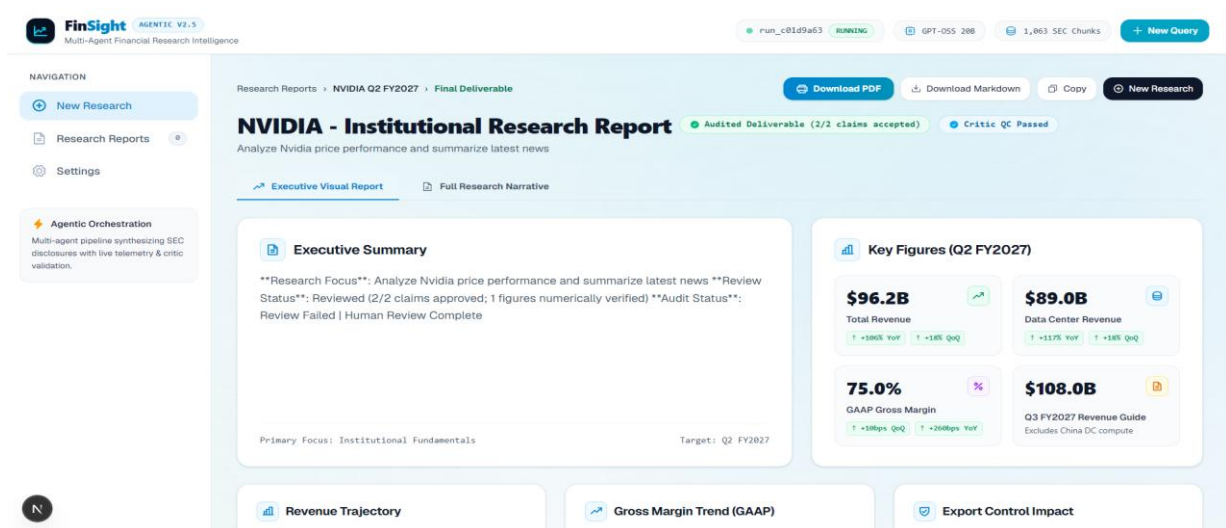


Figure 10.4. Audited Research Report

B. Source-Code Repository

The FinSight source code is organized into separate backend, frontend, evaluation, documentation, and configuration components. The backend contains the FastAPI application, LangGraph workflow, specialized research agents, verification modules, and evaluation suite. The frontend is implemented using Next.js and React. Repository URL: <https://github.com/kishanthraja3/Finsight>

Table I. Major FinSight Software Modules

Module	Responsibility
main.py	FastAPI application and REST/SSE endpoints
config.py	Configuration, model settings, token budgets and limits
planner.py	Research intent classification and task decomposition
financial_research.py	SEC document retrieval using ChromaDB

Module	Responsibility
market_data.py	Market price and valuation data retrieval
news_research.py	Financial news retrieval and relevance filtering
risk.py	Regulatory and legal risk extraction
synthesis.py	Evidence-grounded finding and report generation
verifier.py	Numerical, percentage and temporal validation
critic.py	Task coverage and quality evaluation
replanner.py	Targeted gap-filling for incomplete research tasks
build_graph.py	LangGraph workflow and conditional routing
frontend/	Next.js user interface and research dashboards
backend/eval/	Automated testing and evaluation suite

The repository also contains benchmark data, regression results, documentation, and automated evaluation scripts.

XI. EXPERIMENTAL EVALUATION AND RESULTS

A. Evaluation Metrics

FinSight was evaluated using automated unit tests, integration tests, routing tests, regression tests, recorded execution telemetry, and a comparative benchmark. The evaluation focused on routing correctness, numerical verification, news relevance, citation accuracy, hallucination rate, and system reliability.

Table II. Evaluation Metrics

Metric	FinSight Result
Core regression test pass rate	100% (23/23)
Expanded automated test pass rate	100% (49/49)
Routing intent accuracy	100% (5/5)
News relevance test success	100% (6/6)
Numerical verification tests	100% (12/12)
Citation accuracy	90.0%
Basic RAG citation accuracy	67.5%
No-RAG citation accuracy	0.0%
Hallucination rate	10.0%
Basic RAG hallucination rate	32.5%
No-RAG hallucination rate	42.5%

The automated test suite achieved a complete pass rate across the evaluated test cases. However, this should not be interpreted as 100% factual accuracy of the complete system, since the comparative benchmark measured a 10% hallucination rate for the evaluated Agentic RAG configuration.

B. Testing Methodology

FinSight was evaluated using a multi-level testing methodology covering individual components as well as the complete agentic workflow.

The testing process consisted of the following stages:

- 1. Intent and Routing Testing**

Queries were classified to determine whether SEC retrieval, market data, and news retrieval were required.

- 2. Financial Retrieval Testing**

SEC documents stored in the ChromaDB vector database were retrieved and evaluated using company and section-aware filtering.

- 3. Market and News Testing**

Market-data and news retrieval components were tested independently. News articles were additionally evaluated using the relevance-filtering mechanism.

- 4. Numerical Verification Testing**

Financial figures were tested for unit consistency, rounding differences, and year-over-year calculations.

- 5. Synthesis and Critic Testing**

Generated findings were checked against planned tasks and evidence requirements. The critic was also tested for incomplete multi-period findings and unsupported causal statements.

- 6. Human-in-the-Loop Testing**

Approval, editing, rejection, and revalidation workflows were tested to ensure that analyst decisions were correctly incorporated into the final report.

- 7. End-to-End Regression Testing**

Complete research queries were executed for Apple, NVIDIA, and Microsoft to evaluate the integrated system.

Routing Test Results

Five standardized query types were used to validate conditional routing:

- Microsoft revenue growth → SEC retrieval only
- NVIDIA current price performance → market data only
- Microsoft price performance and risks → SEC + market + news
- Apple European regulatory risks → SEC retrieval only

- Latest NVIDIA news → news retrieval only

All five routing scenarios passed their expected routing conditions.

News Relevance Testing

The news-relevance tests evaluated whether portfolio-related filings and incidental ticker mentions were incorrectly treated as substantive company news. Six test cases were executed, and all six passed. Irrelevant articles were assigned a relevance score of zero and rejected by the verification and critic stages.

Numerical Verification Testing

The numerical verification module was tested using values represented in different scales, including billions and millions, together with percentage rounding variations. The system successfully normalized equivalent representations and rejected unsupported numerical values across all 12 test cases.

C. Performance Results

Three representative end-to-end regression runs were conducted using Apple, NVIDIA, and Microsoft research queries.

Table III. End-to-End Performance Results

Test	Execution Time	LLM Calls	Result	Audit Status
T1 – Apple	~22.4 s	13	Completed	Reviewed — Evidence Limitations Remain
T2 – NVIDIA	~24.1 s	13	Completed	Reviewed — Evidence Limitations Remain
T3 – Microsoft	~21.8 s	13	Completed	Reviewed — Evidence Limitations Remain

The Apple run produced one approved claim and one mathematically verified figure. The NVIDIA run produced two approved claims and successfully performed JSON repair for a truncated model response. The Microsoft run produced two approved claims while correctly separating operating margin from gross margin.

The evidence limitations shown in the final audit status were not treated as system failures. They resulted from missing historical information in the locally indexed corpus. Instead of fabricating unavailable information, FinSight terminated the corresponding gap-fill attempts and explicitly reported the evidence limitation.

D. Success-Metric Achievement

Table IV. Achievement of Project Success Criteria

Success Criterion	Observed Outcome	Status
Intelligent source routing	5/5 routing scenarios passed	Achieved
Numerical verification	12/12 verification tests passed	Achieved
Evidence-grounded synthesis	90% citation accuracy in benchmark	Achieved
News relevance filtering	6/6 relevance tests passed	Achieved
Causal attribution protection	Unsupported causal statements successfully separated	Achieved
Fail-closed auditability	Evidence limitations explicitly recorded	Achieved
Human-in-the-loop governance	Approval, editing and rejection workflows tested	Achieved
Multi-model fallback	Fallback mechanism successfully tested	Achieved

These outcomes demonstrate that the major functional and governance objectives of FinSight were implemented and validated through automated testing and empirical execution.

XII. COMPARISON, INNOVATION AND SCALABILITY

A. Comparison with Baseline

FinSight was compared against two baseline configurations: **No-RAG (Zero-Shot LLM)** and **Basic RAG**. The comparison used curated financial research questions and evaluated citation accuracy, hallucination rate, reasoning quality, retrieval calls, and retrieved evidence.

Table V. Quantitative Comparison with Baselines

Metric	No-RAG	Basic RAG	FinSight	Improvement
Citation Accuracy	0.0%	67.5%	90.0%	+22.5 percentage points vs Basic RAG
Hallucination Rate	42.5%	32.5%	10.0%	22.5 percentage-point reduction vs Basic RAG
Reasoning Quality	2.38/5	2.12/5	2.50/5	+0.38 vs Basic RAG
Average Retrieval Calls	0.0	1.0	1.0	—
Average Chunks Retrieved	0.0	3.0	6.0	+3 relevant chunks

The results indicate that FinSight's agentic retrieval and verification architecture improved citation accuracy and reduced hallucination rate compared with the evaluated baselines. The benchmark also identified cross-company retrieval contamination in Basic RAG, while FinSight applied company-specific metadata filtering to reduce such contamination.

Qualitative Comparison

Capability	LLM-Only	Basic RAG	FinSight
Query decomposition	Limited	Basic	Deterministic clause decomposition
Financial retrieval	Internal model knowledge	Dense retrieval	Section-aware Agentic RAG
Numerical validation	Limited	Retrieval-based	Deterministic verification
Causal claim handling	Uncontrolled	Context-based	Causal claim splitting
News filtering	Limited	Keyword similarity	Relevance evaluation
Quality control	Single-pass	Basic	Fail-closed critic
Human oversight	No	Limited	Interactive HITL

B. Advantages of the Proposed System

The major advantages of FinSight are as follows:

1. Deterministic-Agentic Hybrid Execution

FinSight uses language models for semantic reasoning and synthesis while performing operations such as query routing, numerical calculations, unit conversion, causal-claim separation, and coverage checking through deterministic logic. This reduces unnecessary dependence on LLM reasoning for operations that can be explicitly validated.

2. Reduced Cross-Company Contamination

Company metadata is incorporated into the retrieval process so that evidence belonging to one company is not incorrectly used for another company's analysis. This is particularly important for financial documents containing similar terminology across companies.

3. Fail-Closed Auditability

When sufficient evidence is unavailable, FinSight does not automatically generate a best-guess result. Instead, the critic records the missing evidence and marks the corresponding limitation in the final report.

4. Surgical Replanning

When a particular research task remains incomplete, the replanner targets the missing task rather than restarting the entire research workflow. This preserves previously verified findings.

5. Human Governance with Revalidation

FinSight incorporates human review into the workflow. Analysts can approve, edit, or reject findings, and edited findings can be revalidated before final publication.

C. Innovation

The major technical innovations of FinSight are:

1. Deterministic Causal Attribution Splitting

FinSight separates verified financial figures from unsupported causal statements. When a generated finding combines a verified number with an unverified explanation, the verifier isolates the numerical claim and prevents the unsupported causal statement from being presented as fact. This functionality was validated through the causal-claim testing suite.

2. Multi-Tier News Relevance Filtering

Instead of directly summarizing every article returned by a financial news API, FinSight evaluates the relevance of each article using company and ticker context and identifies portfolio-related or incidental mentions. The approach was validated through six news-relevance tests.

3. Fail-Closed Coverage Ledger

FinSight maintains a coverage ledger connecting planned research tasks with retrieved evidence, verification results, and limitations. If required evidence cannot be obtained after bounded retries, the system records the limitation instead of generating unsupported information

D. Scalability

FinSight has been designed with a modular architecture in which individual agents can be extended or replaced without redesigning the complete workflow. The current implementation includes separate modules for planning, financial retrieval, market data, news, risk analysis, synthesis, verification, criticism, and replanning.

The system also supports multiple LLM models through a centralized client with fallback handling. Conditional routing prevents unnecessary external API calls, while per-run limits prevent uncontrolled execution loops and token consumption.

Current Scalability

- Modular LangGraph agent architecture.
- Conditional execution of research components.

- Multi-model fallback mechanism.
- Per-run LLM call and token limits.
- Local ChromaDB vector retrieval.
- Separate frontend and backend services.

Future Scalability

For large-scale deployment, the current local ChromaDB store can be migrated to a distributed vector database. LangGraph's in-memory checkpointing can be replaced by persistent distributed storage such as PostgreSQL or Redis. Retrieval tasks can also be executed concurrently for multiple companies or financial segments.

E. Real-World Applicability

FinSight can support several practical financial research activities.

1. Equity Research Decision Support

The system can assist analysts in extracting financial figures, calculating growth trends, reviewing management disclosures, and compiling evidence-backed research findings.

2. Regulatory and Compliance Research

FinSight can assist in reviewing regulatory and legal disclosures, including SEC Item 1A risk factors and Item 3 legal proceedings, to identify relevant risk information.

3. Credit and Risk Analysis

The combination of market information, financial disclosures, and risk findings can support preliminary analysis of financial and operational risks.

However, FinSight is intended as an analyst-support system rather than an autonomous financial advisory system. Its results depend on the available document corpus and external market/news APIs, and important financial decisions still require human review.

XIII. CONCLUSION AND FUTURE WORK

A. Conclusion

FinSight is an **Agentic Financial Research and Analysis System** designed to improve the reliability, traceability, and efficiency of financial research. The system combines specialized research agents, intelligent source routing, SEC-based retrieval, market-data and news analysis, evidence-grounded synthesis,

deterministic numerical verification, critic-based quality control, targeted replanning, and human-in-the-loop auditing.

The experimental evaluation demonstrated a **100% pass rate across 49 automated tests**, including routing, numerical verification, news relevance, and workflow testing. The comparative benchmark achieved **90.0% citation accuracy** and a **10.0% hallucination rate**, compared with 67.5% citation accuracy and 32.5% hallucination rate for the Basic RAG baseline.

The system was also evaluated through end-to-end Apple, NVIDIA, and Microsoft research runs. These experiments demonstrated successful multi-agent execution, evidence verification, report generation, and human-audit functionality. Importantly, FinSight follows a fail-closed approach: when the required evidence is unavailable, it reports an evidence limitation rather than fabricating information.

Overall, the results demonstrate that combining agentic task decomposition with retrieval, deterministic verification, quality control, and human oversight can provide a more auditable financial research workflow than conventional single-pass LLM or basic RAG approaches.

B. Future Work

The following improvements can further enhance FinSight:

- 1. Expansion of the Financial Corpus**

Extend the current SEC document corpus beyond the currently indexed companies and filings to support a larger range of public companies and reporting periods.

- 2. Live SEC Data Integration**

Integrate live SEC EDGAR retrieval so that newly published filings can be accessed without requiring prior local ingestion.

- 3. Distributed Vector Database**

Replace the local ChromaDB deployment with a distributed vector database for large-scale financial document retrieval.

- 4. Persistent Distributed Checkpointing**

Replace in-memory workflow checkpointing with persistent storage such as PostgreSQL or Redis to support multi-user deployments.

- 5. Parallel Research Execution**

Introduce parallel retrieval for independent financial, market, and news tasks to reduce end-to-end execution time.

6. **Broader Benchmarking**

Expand the evaluation dataset with more companies, financial questions, market conditions, and reporting periods to obtain more statistically representative performance measurements.

7. **Improved News Coverage**

Integrate additional high-quality financial news sources and further improve relevance ranking to distinguish substantive corporate developments from incidental financial filings.

8. **Cloud Deployment**

Deploy the frontend, backend, vector database, and supporting services on cloud infrastructure for multi-user access and scalable execution.

9. **Advanced Financial Analytics**

Extend the system with additional valuation analysis, peer comparison, financial forecasting, and portfolio-level risk analysis.

10. **Enhanced Human-in-the-Loop Controls**

Expand analyst review capabilities with richer editing, evidence comparison, confidence explanations, and configurable approval policies.

These future improvements would allow FinSight to move from a validated academic prototype toward a more scalable financial research platform while retaining its emphasis on evidence, verification, transparency, and human oversight.

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